

Project 5: LangGraph Demand Sensing & Supply Chain Command Center Copilot

A C5i-style project brief for building an agentic supply chain command-center copilot using LangGraph, demand sensing signals, tools, routing, critique loops, and human approval.

Section	Details
Project	Project 5: LangGraph Demand Sensing & Supply Chain Command Center Copilot
Business Themes	Supply Chain Analytics; Demand Sensing; Inventory Management; Supply Chain Command Center; Real-Time Visibility; Waste Reduction
Technology Stack / Themes	Agentic AI with LangGraph; state graph routing, edge routing, tool execution nodes, critique loops, human-in-the-loop approval, demand sensing analytics, what-if simulation
Project Description	Build a LangGraph-based command-center copilot for a CPG/retail supply chain team. The system should detect demand spikes or drops, explain drivers using internal and external signals, recommend inventory or replenishment actions, and escalate decisions requiring human approval.
High-Level Work Areas	1. Create state schema for demand signals, inventory positions, service levels, exceptions, root-cause hypotheses, recommendations, and approval status. 2. Build tools for demand data lookup, inventory lookup, promotion calendar, weather/event signals, supplier delay data, and what-if simulations. 3. Implement LangGraph nodes for detect, diagnose, retrieve evidence, recommend, critique, and escalate. 4. Route cases by severity, confidence, and business impact. 5. Add HITL step for replenishment or inventory transfer decisions. 6. Build a dashboard-style demo with alerts, evidence, and action recommendations. 7. Evaluate alert precision, recommendation correctness, and trace quality.
Synthetic / Input Data	Synthetic SKU-store-week demand data, inventory snapshots, promotion calendar, supplier delay logs, weather/event data, shipment data, service-level targets
Evaluation Metrics	Demand-signal detection accuracy, stockout/overstock alert precision, recommendation quality, reduction in false alerts, explainability, HITL approval rate, latency and reliability
Progressive Demo Sequence	1. Start with a command-center question: "Which SKUs are at risk of stockout next week and why?" 2. Show a basic forecast or threshold rule. Highlight that it detects anomalies but does not explain root cause or recommend action. 3. Add a simple LLM summary of demand and inventory tables. Highlight that table reasoning is fragile without tools and formulas. 4. Add tool calls for demand, inventory, promotion calendar, and supplier status. Ask: "Why did SKU X demand spike in West region?" 5. Add LangGraph routing: low-risk alerts go to summary, high-risk alerts go to diagnosis and HITL approval. 6. Add root-cause node: promotion effect, weather/event signal, supplier delay, competitor activity, or data anomaly. 7. Add recommendation node: transfer inventory, expedite supplier order, reduce promotion intensity, or wait-and-watch. 8. Add critique node to check whether the recommendation is supported by evidence and whether constraints are violated. 9. Add human approval step for high-cost actions; demonstrate approve/reject/edit flow. 10. Final limitations to highlight: demand sensing quality depends on data freshness, external signals may be noisy, what-if simulations are simplified, and recommendations need ERP/SCM integration before production. 11. Future scope: real-time streaming, optimization solver integration, digital twin, forecast model benchmarking, supplier risk scoring, and automated command-center alerts.